
LATENT PLANNING VIA EMBEDDING ARITHMETIC: A CONTRASTIVE APPROACH TO STRATEGIC REASONING

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ABSTRACT

Planning in high-dimensional decision spaces is increasingly being examined through the lens of learned representations, a shift that reframes the problem of action selection as one of geometric organization rather than explicit rollout. In contrast to paradigms predicated on the training of policies or value heads, we interrogate whether planning can be conducted directly within an evaluation-aligned embedding space, thereby dispensing with the auxiliary machinery those paradigms ordinarily presuppose. To this end we introduce SOLIS, which induces such a space by means of supervised contrastive learning. Within the resulting representation, similarity of outcome is encoded as spatial proximity, while a single global *advantage vector* orients the space along the continuum extending from losing to winning regions. Candidate actions are subsequently ranked according to the degree of their alignment with this direction, a formulation that reduces planning to elementary vector operations in latent space. We instantiate the approach in chess, wherein SOLIS attains competitive strength under constrained conditions while relying on nothing more than a shallow search steered by the learned embedding. More broadly, our findings intimate that evaluation-aligned latent planning constitutes a lightweight alternative to conventional dynamics models and to policy learning alike.

1 INTRODUCTION

Planning in high-dimensional state spaces persists as a central challenge in artificial intelligence. Conventional reinforcement learning methods, which rest upon explicit search or upon learned policies, are liable to become computationally unwieldy as the space of admissible actions expands (Jiang et al., 2023; Giesemann and Pokorný, 2022). A recent and increasingly influential line of inquiry has consequently turned to latent planning, an approach that reasons over compact latent representations in lieu of raw states, thereby attenuating combinatorial complexity while affording decision-making of a more human-like character (Hafner et al., 2019; 2020; Schrittwieser et al., 2020).

We investigate whether a learned embedding space can, of itself, underwrite planning once decision-making is recast as interpolation between regions of differential advantage. To this end we introduce **SOLIS** (*Strategic Optimization via Latent Interpolative Spaces*), a chess engine erected upon a transformer encoder that is trained with supervised contrastive learning so as to embed states into an evaluation-aligned latent space. Within this space, distance correlates with similarity of outcome, and directional displacement corresponds, heuristically, to progression toward favorable states. SOLIS exploits precisely this structure, ranking candidate moves by the extent to which they align with a single global “advantage vector” that orients the latent space from losing regions toward winning ones.

Even under markedly modest search budgets, SOLIS yields strong results. Confined to a mere 5-ply search under 50 ms/move blitz conditions, it attains an Elo of 2500+; and in equal-depth encounters at search depth $S \in \{3, 4, 5\}$ with a fixed branching factor, it proves competitive with a configured Stockfish baseline within our experimental setting. Visualizations of authentic games furnish an additional line of evidence, demonstrating that game trajectories trace smooth paths through the embedding space and thereby lend the method a measure of qualitative interpretability.

In contradistinction to latent planning methods that couple learned dynamics with policy or value heads (Hafner et al., 2019; 2020; Schrittwieser et al., 2020), SOLIS predicates action selection exclusively upon the representation afforded by an evaluation-aligned space, without recourse to any such auxiliary head. Although chess serves here as our testbed, the vector-based latent planning approach is, in principle, extensible to other perfect-information domains in which state evaluations are either available or learnable.

2 BACKGROUND

Planning in very large action spaces presents a formidable challenge in artificial intelligence, inasmuch as the number of admissible action sequences proliferates exponentially with every additional decision step. This combinatorial explosion demands careful modeling and a considered, critical deliberation over the construction of a robust methodology. The present section accordingly expounds the salient details governing the operation of our proposed model.

2.1 MARKOV DECISION PROCESSES

Sequential decision-making tasks are canonically formalized as Markov Decision Processes (MDPs), each expressible as a tuple $\{\mathcal{S}, \mathcal{A}, T, R, p(s_0), \gamma\}$, in which $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, T the transition model, R the reward function, $p(s_0)$ the initial state distribution, and $\gamma \in [0, 1]$ the discount factor attenuating future rewards (Puterman, 2014). A policy is correspondingly defined as $\pi : \mathcal{S} \rightarrow p(\mathcal{A})$, namely a mapping from states to probability distributions over actions.

Within this setting, a *model* denotes reversible access to the MDP dynamics: presented with a state–action pair, the model predicts the ensuing state together with its associated reward (Moerland et al., 2022). For games such as chess and Go, the rules of play themselves constitute a direct and exact model. In domains where such rules are not antecedently known, by contrast, the model must be approximated inductively from data (Giesemann and Pokorný, 2022).

Planning and reinforcement learning may then be distinguished precisely by the manner in which each exploits this access to MDP dynamics (Moerland et al., 2022). Planning methods presuppose access to a model and simulate forward from the current state in order to select an action, in a manner reminiscent of the deliberation humans undertake mentally (Russell and Norvig, 1995). Because the solution so obtained is indexed to the current state, it is regarded as *local* and is accordingly discarded once execution has occurred. Reinforcement learning methods, by contrast, characteristically lack reversible access to the MDP dynamics (Sutton and Barto, 1998); obliged to interact with the environment incrementally, step by step, they instead acquire a *global solution*—a policy or value function, say—that generalizes across the entirety of the state space.

Model-based reinforcement learning conjoins both elements, availing itself of a model while concurrently learning a global solution. According to Moerland et al. (2022), this synthesis admits of two distinct forms. The first is model-based RL with a learned model (e.g., Dyna (Sutton, 1991)), wherein both the dynamics and the global solution are learned in tandem. The second is model-based RL with a known model (e.g., AlphaZero (Silver et al., 2017)), wherein the dynamics are furnished a priori and only the global solution remains to be learned.

2.1.1 MODEL-BASED RL WITH A LEARNED MODEL

An extensive body of work subsumes itself under the category of model-based RL with a learned model, in which both the dynamics and the global solution are jointly acquired. PlaNet (Hafner et al., 2019) learns a latent dynamics model and marries it to a learned global solution. Dreamer (Hafner et al., 2020), together with its successors DreamerV2 (Hafner et al., 2022) and DreamerV3 (Hafner et al., 2024), extends this paradigm by scaling to progressively more intricate tasks, all the while retaining global value functions and policies. MuZero (Schrittwieser et al., 2020) likewise learns a latent dynamics model, yet integrates it with tree search so as to fortify planning even as it trains a global policy and value function. Further exemplars include Dyna (Sutton, 1991) and MBPO (Janner et al., 2021), each of which enlists learned models in the service of value or policy learning.

2.1.2 MODEL-BASED RL WITH A KNOWN MODEL

When the environment dynamics are available a priori, the algorithm is at liberty to concentrate upon the coupling of search with learning. AlphaZero (Silver et al., 2017) adopts the exact game rules as its model, conjoining Monte Carlo tree search (MCTS) with learned policy and value heads. SAVE (Hamrick et al., 2020) analogously presupposes a simulator and integrates Q-learning with MCTS: a learned Q-function operates as a prior that guides the search, while the Q-estimates the search in turn produces are amortized back into the network.

2.1.3 LATENT PLANNING

Latent planning simplifies decision-making by operating within a compressed representation space in preference to the raw state space. PlaNet (Hafner et al., 2019), Dreamer (Hafner et al., 2020), and MuZero (Schrittwieser et al., 2020) each pair latent dynamics models with value or policy heads, thereby enabling planning through simulations conducted in the learned space. Expansive Latent Space Trees (ELAST) (Giesemann and Pokorný, 2022) recast latent planning as explorative tree search, conjoining contrastive representations with sampling-based expansion in order to confront long-horizon control. Trajectory Autoencoding Planner (TAP) (Jiang et al., 2023) pursues a cognate strategy, learning a compact latent action space and planning over discrete action codes.

SOLIS, by contrast, neither trains policies or value heads nor learns a dynamics model; it instead distills the evaluation targets directly into the representation itself. Planning thereafter proceeds through movement between regions aligned with favorable outcomes, drawing solely upon the representation space at inference time.

2.2 CHESS ENGINES

IBM decisively scaled chess computation in 1997, its Deep Blue engine defeating the reigning world chess champion, Garry Kasparov, in a head-to-head match. Early incarnations of Stockfish (Romstad et al., 2008) adhered to a methodology closely paralleling that of Deep Blue: handcrafted evaluation functions coupled with deep alpha-beta tree search. Compelling though this approach was—and superhuman though the performance it achieved—it became increasingly apparent that the operative bottleneck lay in the aggregate chess mastery of those charged with engineering the static evaluation function.

The rating bottlenecks attendant upon static evaluation galvanized a vein of research emancipated from human guidance. AlphaZero (Silver et al., 2017), together with its open-source counterpart Leela Chess Zero (Pascutto and Linscott, 2018), effected a substantial leap toward engines that learned to evaluate positions wholly through self-play reinforcement learning, unencumbered by any prior knowledge beyond the rules of chess. In a parallel development, modern versions of Stockfish embraced an efficiently updatable neural network evaluation module (NNUE) (Nasu, 2018), expediting position evaluation while nonetheless preserving traditional alpha-beta search.

While these advances have yielded engines that far outstrip human players in strength, the underlying search algorithms have remained largely unaltered. They continue to abide by principles first articulated by v. Neumann (1928) and Shannon (1950)—principles subsequently incorporated into Deep Blue—predicated upon deep alpha-beta search through millions of states (Lai, 2015). In the present work we interrogate whether contrastive learning can engender a more efficient search process by construing planning as traversal through a high-dimensional embedding space.

2.3 CONTRASTIVE LEARNING

Contrastive representation learning induces a model to embed similar inputs in closer proximity and dissimilar inputs at greater remove within a latent space (Chen et al., 2020). More concretely, given a batch I of training examples encompassing an anchor input x_i , an encoder $f(\cdot)$ maps the anchor to a normalized embedding $z_i = f(x_i)$. Positive pairs (z_i, z_j) correspond to semantically kindred inputs, whereas all remaining pairs of samples within the batch are apprehended as negatives.

A common contrastive objective is InfoNCE (van den Oord et al., 2019), defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

where $A(i) = I \setminus \{i\}$ designates the set of all samples in the batch save the anchor i , $\text{sim}(\cdot, \cdot)$ denotes a similarity metric, and τ constitutes a scalar temperature parameter. In the self-supervised regime of contrastive learning, the positive sample z_j is furnished as an augmentation of the anchor z_i , whereby labels are rendered unnecessary (Chen et al., 2020; Khosla et al., 2021).

When labels are available, the supervised contrastive loss (SupCon) (Khosla et al., 2021) generalizes InfoNCE by admitting multiple positive examples per anchor. The SupCon loss is defined as:

$$\mathcal{L}_{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)}, \quad (2)$$

where $P(i)$ denotes the set of positive examples corresponding to anchor i , and $A(i)$ retains the definition stipulated above. An augmentation in the number of positive and negative pairs has been demonstrated to enhance the quality of the learned representation (Khosla et al., 2021; Chowdhury et al., 2024).

3 METHODS

We now delineate the constituent components of our approach, encompassing tokenization, model architecture, the training procedure, and action selection at inference time.

3.1 INPUT REPRESENTATION AND TOKENIZATION

Each chess position is represented by its Forsyth-Edwards Notation (FEN) ASCII string, which compactly encodes the piece placement, the side to move, castling rights, the en passant target, the half-move clock, and the full-move counter. Adhering to the tokenization scheme advanced by Ruoss et al. (2024), we tokenize each FEN into a fixed-length sequence of 77 tokens by expanding its run-length encodings. As Ruoss et al. (2024) observe, this tokenization renders chess a non-Markovian problem, inasmuch as information pertaining to past states is elided even though it may prove consequential for future actions in certain special cases (e.g., threefold repetition or the fifty-move rule). We revisit this consideration at greater length in Section 5.

3.2 ENCODER ARCHITECTURE

Our encoder is a multi-layer transformer (Vaswani et al., 2023), adapted for the tokenized chess positions at hand. We train two variants thereof, SOLIS Base and SOLIS Mini, and summarize the model configurations employed throughout our experiments in Table 1.

Table 1: Transformer model configurations adopted across our experiments.

Model	Layers	Embedding dim. D	Heads	MLP Size	Params
SOLIS Mini	6	128	8	256	0.8M
SOLIS Base	6	1024	16	1024	41M

Each input sequence is embedded into D -dimensional vectors by way of a learned token embedding matrix. A special classification token (CLS) (Devlin et al., 2019; Dosovitskiy et al., 2021) is prepended to the embedded sequence in order to aggregate information across the whole of the input. Because the input sequences are of fixed length, we superimpose a learned positional encoding upon each token embedding (Ruoss et al., 2024). The resulting sequence is thereafter processed by stacked transformer encoder layers endowed with GELU (Hendrycks and Gimpel, 2023) activations and dropout (Srivastava et al., 2014). The terminal hidden state corresponding to the CLS token is extracted, passed through a linear projection, and finally ℓ_2 normalized.

3.3 SUPERVISED CONTRASTIVE TRAINING

We train our encoder upon 5 million randomly sampled positions drawn from the ChessBench dataset (Ruoss et al., 2024), each annotated with a Stockfish-evaluated win probability for the player to move. All evaluations are normalized so as to represent White’s perspective, such that values approaching 1.0 betoken a decisive advantage for White while values approaching 0.0 signal a corresponding advantage for Black.

To delimit the positive samples for training, we designate δ as the evaluation margin governing the identification of similar positions. For each anchor, we precompute the set of all positions whose win probabilities differ by less than δ and, during training, randomly sample five thereof as positives. To construct the batch-level mask, we compare the win probabilities of all samples and mark a pair as positive whenever their evaluation difference falls below δ . This margin-based sampling scheme draws together states of nearly commensurate evaluation while forcing apart positions separated by larger gaps. In virtue of so defining the positives, the embedding space is, in the limit, induced to instantiate an ordering that runs from losing to winning positions.

We apply supervised contrastive learning (SupCon) (Khosla et al., 2021), employing the constructed batch masks to demarcate positive from negative pairs. Pairwise scores are computed by means of cosine similarity, the temperature $\tau = 0.07$ (Girdhar et al., 2023; Radford et al., 2021) modulating the sharpness of the resulting distribution. Optimization proceeds via stochastic gradient descent with momentum 0.9 (Dosovitskiy et al., 2021; Chen et al., 2020; He et al., 2015), a batch size of 128, over the course of 400,000 steps. Training is conducted on six NVIDIA L40S GPUs (48 GB apiece) under data parallelism (Paszke et al., 2019). The complete slate of training hyperparameters is furnished in Appendix A.2.

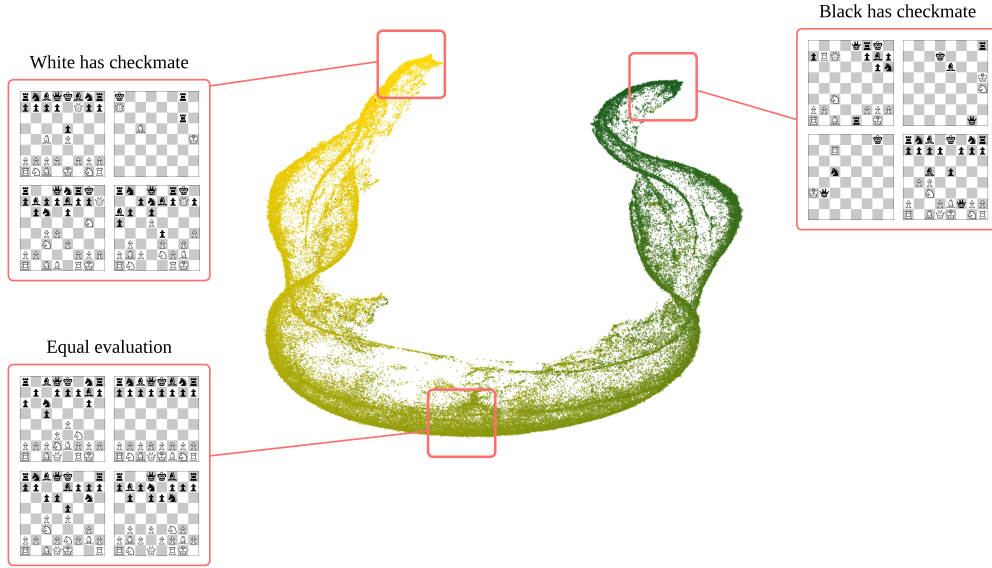


Figure 1: UMAP projection of the learned embedding space of our Base model, colored according to win probability (gold denotes White favored, green denotes Black favored).

We visualize the learned representation space of our Base model in Figure 1, availing ourselves of UMAP (McInnes et al., 2020) for the 2D projection. Each point corresponds to an encoded chess position drawn from a test dataset and is colored according to its Stockfish-evaluated win probability for White. Positions in which White commands a decisive advantage are rendered in gold (*White has checkmate*), whereas those in which Black commands the corresponding advantage are rendered in dark green (*Black has checkmate*).

3.4 EMBEDDING-GUIDED SEARCH

At inference we plan directly within the learned space, ranking the legal continuations by reference to a single global advantage direction. To this end we compute means from a held-out dataset at the extremes of evaluation,

$$\mu_{\text{White}} = \mathbb{E}[z \mid p=1.0], \quad \mu_{\text{Black}} = \mathbb{E}[z \mid p=0.0],$$

and thereupon define the normalized advantage vector

$$\vec{a} = \frac{\mu_{\text{White}} - \mu_{\text{Black}}}{\|\mu_{\text{White}} - \mu_{\text{Black}}\|}.$$

In our experiments we contemplate two distinct mechanisms whereby this vector may be converted into scalar scores.

3.4.1 UNANCHORED (GLOBAL-DIRECTIONAL)

The first approach scores each child embedding directly according to its alignment with the advantage direction. Each child embedding z' is thus assigned the score

$$\text{score}_{\text{dir}}(z') = \langle z', \vec{a} \rangle,$$

thereby favoring those moves whose embeddings extend furthest along the advantage direction, irrespective of the origin. It is this scoring mechanism that Figure 2 depicts by way of illustration.

3.4.2 ANCHORED (RELATIVE PROJECTION)

Alternatively, we may shift by the Black mean and score

$$\text{score}_{\text{anch}}(z') = \langle z' - \mu_{\text{Black}}, \vec{a} \rangle,$$

whereby the value becomes proportional to the signed distance from μ_{Black} along $\vec{a}$. This expedient dispenses with any dependence upon an arbitrary origin and, in practice, furnishes a more stable basis for the comparison of continuations.

The search process (Figure 2) is identical under both scoring mechanisms. Departing from the “Starting board state”, we “Extract possible moves” and embed each child in parallel by means of the “Transformer encoder”. We thereupon compute either $\text{score}_{\text{dir}}$ or $\text{score}_{\text{anch}}$ within the “Cosine similarity” block and retain the “Top- W ” children. Should we have reached “ S levels deep?”, the procedure terminates and we “Select first move. . .”; otherwise we prosecute the “Recursive search” under a min-max preference (White maximizing, Black minimizing). We further enlist Zobrist hashing (Zobrist, 1990) in the service of a transposition table, so as to obviate the re-evaluation of positions already encountered.

4 RESULTS

4.1 RATING CALCULATION

To estimate Elo ratings, each model was pitted against Stockfish under a fixed per-move time limit of 50 ms (Ruoss et al., 2024); the full Stockfish configurations appear in Appendix A.3. We conducted an ablation study spanning both the breadth and the depth of the search. At each depth/width configuration, our models contested 600+ games with alternating colors against Stockfish configured at a range of Elo caps, and both planning processes—unanchored and anchored alike—were evaluated under this common protocol. The caps were so selected as to guarantee that each model configuration yielded at least one matchup exhibiting a positive win rate and at least one exhibiting a negative win rate. Elo ratings, together with their 95% confidence intervals, were estimated by Bayesian logistic regression via the BayesElo (Coulom, 2008) program under the default confidence parameter of 0.5. Elo estimates for the unanchored planning method are reported in Figure 3, while those for the anchored method are displayed in Figure 4. For the sake of Elo reproducibility, the Stockfish rating caps and the detailed match results between SOLIS and Stockfish are consigned to Appendix A.6.

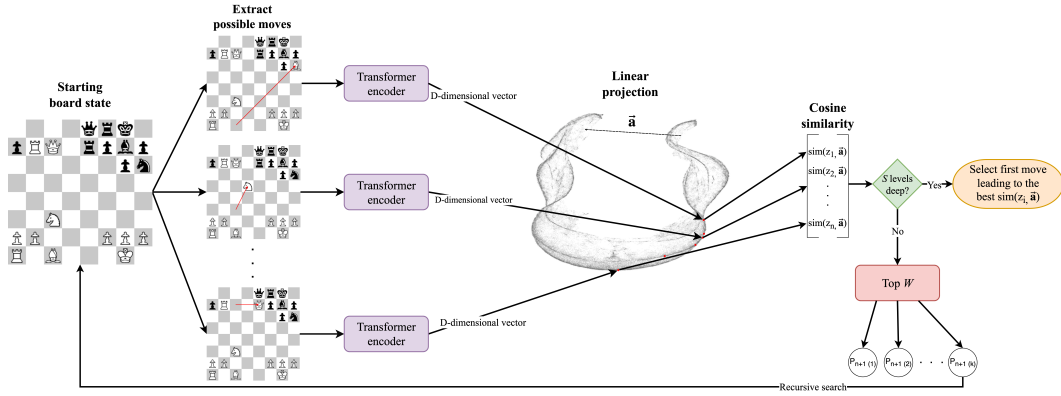


Figure 2: System overview of SOLIS. Candidate moves are embedded and subsequently scored according to their alignment with the global advantage vector. For the sake of simplicity, this diagram portrays the unanchored scoring mechanism $\cos(z', \vec{a})$.

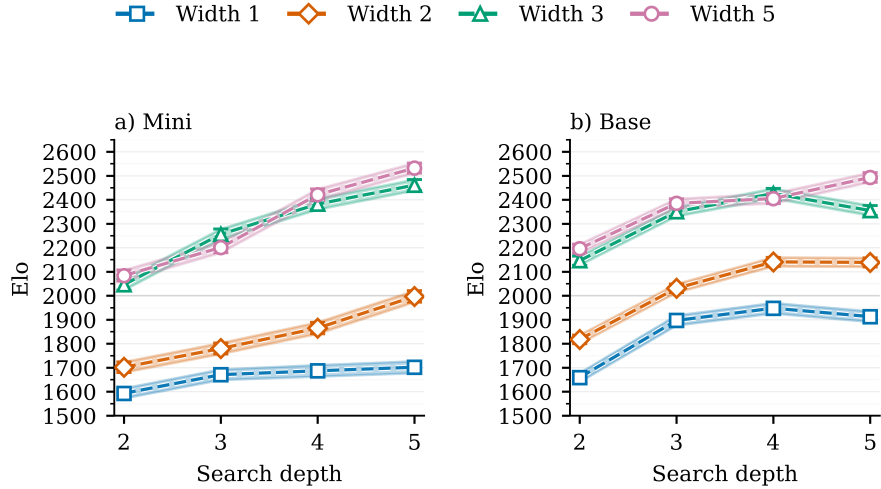


Figure 3: Elo calculations for our Mini and Base models across varying search widths and depths under the *unanchored* planning method. The shaded bands demarcate 95% confidence intervals.

Unanchored ratings Under unanchored planning, Elo ratings scale consistently with the extent of search. Depth confers the most considerable gains, whereas width exhibits a pronounced discontinuity in ascending from $W \in \{1, 2\}$ to $W=3$ yet yields scant further improvement thereafter at $W=5$. This plateau intimates that the strongest continuations reside, as a rule, among the top three candidates, so that broader expansion contributes little additional strength. At shallow search depths, SOLIS Base surpasses Mini by roughly 100–250 Elo; at depths 4 and 5, however, Mini attains parity with Base and indeed overtakes it at $W \in \{3, 5\}$ under this planning mechanism.

Anchored ratings Under anchored scoring, Elo estimates ascend smoothly with both depth and width, and the disparity between Base and Mini persists within the 100–250 range. Both models eclipse 2500 Elo at depth 5; yet, in contradistinction to the unanchored case, the Base model proves consistently stronger than Mini across the entirety of the settings examined. Anchoring likewise confers a steady 10–30 Elo gain upon Base, which suggests that shifting relative to the Black mean affords a more dependable basis for evaluation.

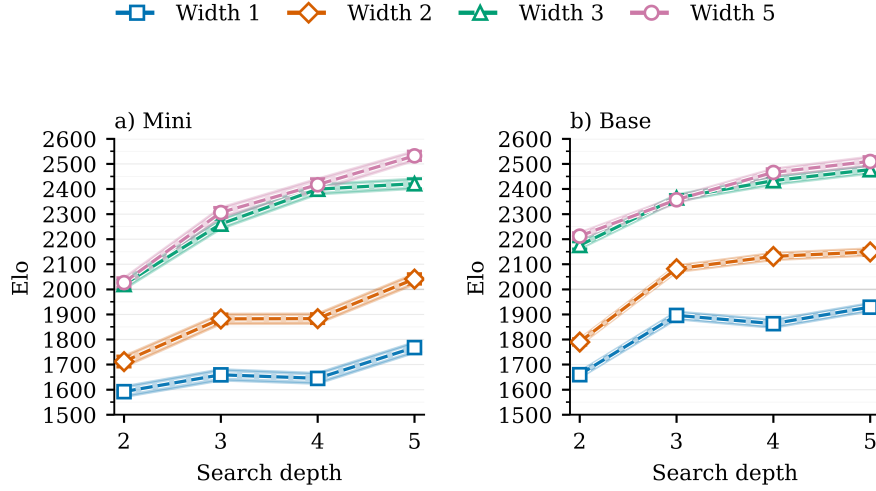


Figure 4: Elo calculations for our Mini and Base models across varying search widths and depths under the *anchored* planning method. The shaded bands demarcate 95% confidence intervals.

4.2 EQUAL-DEPTH SIMULATIONS

We evaluate SOLIS in head-to-head matches against Stockfish at fixed search depths. Guided by the Elo scaling experiments of Section 4.1, we fix the branching factor at $W=3$, since wider searches disclosed only marginal gains beyond this threshold—a finding consonant with the observation that the strongest continuations generally reside among the top three candidates. Stockfish, by contrast, retains its default expansion, unconstrained by any fixed branching factor. At each depth $S \in \{3, 4, 5\}$ we simulate 200 games with alternating colors. The outcomes for unanchored and anchored scoring are tabulated in Table 2.

Simulation results Under unanchored scoring, SOLIS Base accrues higher match points than Stockfish across every depth tested, whereas Mini performs comparably at depths 3 and 5 yet trails at depth 4. Under anchored scoring, Base is narrowly bested at depth 3 but outperforms Stockfish at depths 4 and 5, while Mini secures victories at depths 3 and 5 but once again falls short at depth 4.

Table 2: Match points accrued by SOLIS against Stockfish. The middle portion reports the outcomes obtained when planning proceeds by direct similarity with the global advantage vector, whereas the rightmost portion reports those obtained when planning proceeds by anchored projection. Higher values are preferable; draws are credited as 0.5.

Config	Depth	Direct Similarity		Anchored Projection	
		SOLIS score	Stockfish score	SOLIS score	Stockfish score
Base	3	106	94	98	102
	4	109	91	108	92
	5	107.5	92.5	110	90
Mini	3	108	92	105	95
	4	85	115	91	109
	5	101	99	113	87

Nodes searched For SOLIS operating under a fixed branching factor of $W=3$, the maximal number of nodes searched at depth S is given by $N(S) = \sum_{i=0}^S 3^i$, which yields $N(3) = 40$, $N(4) = 121$, and $N(5) = 364$. Stockfish, by comparison, explores a variable number of nodes per move, contingent upon its pruning and heuristics: at depth 3 it searched an average of 137 nodes (with a maximum of 1625), at depth 4 an average of 214 nodes (maximum 4463), and at depth 5 an

average of 338 nodes (maximum 10611). Notwithstanding its far more parsimonious node budget, SOLIS remains competitive with this Stockfish configuration—a result which intimates that **much of the strength ordinarily accrued through broad expansion can instead be recovered by way of higher-quality representations.**

4.3 INTERPRETABLE GAME TRAJECTORIES

We visualize latent sequences of moves drawn from real games in order to underscore the interpretability of the embedding space. Figure 5 presents three representative exemplars: a game won by White, a draw that remains balanced throughout, and a game won by Black. Each sequence is plotted over the identical 2D projection derived from a sample dataset, the positions being embedded independently and thereafter connected by arrows that mark the progression of movement. Games in which one player prevails decisively tend to trace smooth paths through the space, whereas closely contested games oscillate about the center. These trajectories complement the earlier embedding visualization (Figure 1) by rendering manifest how the planning process is translated into movement toward the regions answering to winning or losing outcomes.

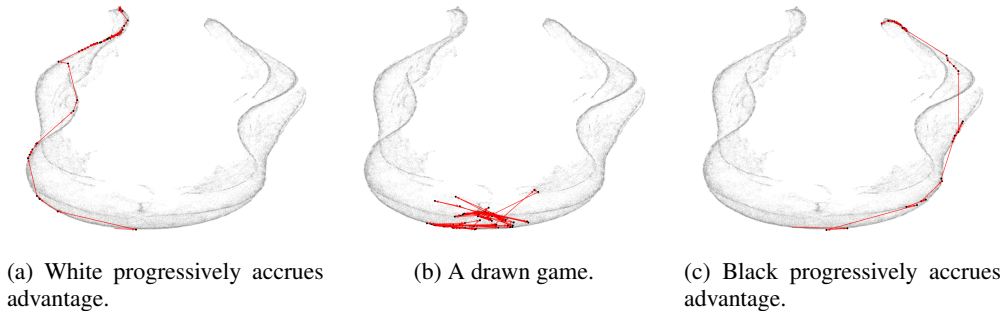


Figure 5: Latent trajectory visualizations of three games embedded within the shared representation space. The red arrows trace the progression of positions as each game unfolds.

5 DISCUSSION

5.1 LIMITATIONS

Blindness to threefold repetition As adumbrated above, SOLIS predicates its moves upon alignment with an advantage vector, thereby achieving effective decision-making at minimal computational overhead. Yet an intrinsic corollary of this parsimonious formulation is the disregard of move history, a disregard that can occasion manifestly irrational behavior. The most conspicuous instance is blindness to threefold repetition: should the same position—inclusive of the side to move together with castling and en-passant rights—recur three times at any junctures in the game, either player may claim a draw. A discerning opponent can accordingly engineer such repetitions (perpetual checks, for example) so as to neutralize winning positions that SOLIS would otherwise convert. Game *C* in Appendix A.4 furnishes an example in which SOLIS attains a winning position against Stockfish yet acquiesces to a premature draw by repetition.

Indecisiveness in winning positions SOLIS is unable to discriminate between forcing positions of differing length. Both a mate-in-1 and a mate-in-20, for instance, carry a win probability of 1.0, notwithstanding that the shorter line ought manifestly to be preferred. When SOLIS elects the more protracted continuation, it affords the opponent latitude to complicate the position, an eventuality that may occasionally precipitate gratuitous draws or even outright losses. Ruoss et al. (2024) confronted this very difficulty and circumvented it by consulting Stockfish as an external oracle. SOLIS, by contrast, elects to progress toward its own solution—though this remains a lacuna yet to be redressed. Game *I* in Appendix A.5 exhibits a case in which SOLIS commands a multiplicity of winning continuations over several consecutive moves, yet commits suboptimal moves that culminate in a draw.

5.2 CONCLUSION

We investigated whether planning can be conducted directly within an evaluation-aligned latent space, in lieu of training a policy or value head. To this end we introduced SOLIS, which learns a contrastive embedding wherein proximity reflects similarity of outcome and a single global *advantage vector* points from losing regions toward winning ones. Planning thereupon reduces to vector arithmetic: candidate actions are ranked according to their alignment with this direction, and a modest search is expanded accordingly.

In chess, SOLIS operates with a fixed branching factor $W=3$ and shallow depths $S \in \{3, 4, 5\}$. Across these settings it matches or exceeds a time-capped Stockfish in several equal-depth matchups, all the while interrogating far fewer nodes per move (40–364 for SOLIS, as against the hundreds to thousands explored by Stockfish at the same depths). Elo ascends with depth; under anchored scoring, both Mini and Base surpass 2500 Elo at depth five, and anchoring confers a consistent 10–30 Elo gain upon Base. Width, by contrast, yields scant benefit beyond three candidates—an observation that suggests the preponderance of strength derives from the quality of the learned space rather than from broad expansion. The embedding additionally affords legible game paths: decisive games advance steadily toward winning regions, whereas draws linger near neutrality.

Taken together, these results vindicate evaluation-aligned latent planning as a simple and compute-efficient alternative to rollout-based search and to explicit dynamics models. We anticipate that the same principle will generalize to other perfect-information games and to control tasks endowed with large action spaces, in which parsimonious low-branching search is at a premium. Avenues for future work include the incorporation of rule-aware state features, the preferential selection of shorter forced wins, the conjunction of SOLIS with policy priors or MCTS, the study of calibration and uncertainty, and the appraisal of transfer beyond the domain of chess.

REPRODUCIBILITY STATEMENT

The complete training setup, encompassing hyperparameters and implementation details alike, is provided in Appendix A.2. Section 3 delineates the encoder architecture, the input representation, the supervised contrastive objective, and the inference procedure, with model configurations summarized in Table 1. Evaluation details—among them the match setup, the Stockfish configurations, and the rating calculations—are set forth in Section 4 and Appendix A.3. Detailed match results are, in turn, furnished in Appendix A.6 for the purpose of Elo verification.

REFERENCES

- T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations, 2020. URL <https://arxiv.org/abs/2002.05709>.
- S. Chowdhury, G. Hamerly, and M. McGarrity. Active learning strategy using contrastive learning and k-means for aquatic invasive species recognition. In *2024 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 848–858, 2024. doi: 10.1109/WACVW60836.2024.00097.
- R. Coulom. Whole-history rating: A bayesian rating system for players of time-varying strength. In *Computers and Games*, volume 5131 of *LNCS*, pages 113–124. Springer, 2008. doi: 10.1007/978-3-540-87608-3_10.
- J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019. URL <https://arxiv.org/abs/1810.04805>.
- A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale, 2021. URL <https://arxiv.org/abs/2010.11929>.
- R. Gieselmann and F. T. Pokorny. Latent planning via expansive tree search. In A. H. Oh, A. Agarwal, D. Belgrave, and K. Cho, editors, *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/forum?id=zSdz5scsnzU>.

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- R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra. Imagebind: One embedding space to bind them all, 2023. URL <https://arxiv.org/abs/2305.05665>.
- D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. Learning latent dynamics for planning from pixels, 2019. URL <https://arxiv.org/abs/1811.04551>.
- D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to control: Learning behaviors by latent imagination, 2020. URL <https://arxiv.org/abs/1912.01603>.
- D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba. Mastering atari with discrete world models, 2022. URL <https://arxiv.org/abs/2010.02193>.
- D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models, 2024. URL <https://arxiv.org/abs/2301.04104>.
- J. B. Hamrick, V. Bapst, A. Sanchez-Gonzalez, T. Pfaff, T. Weber, L. Buesing, and P. W. Battaglia. Combining q-learning and search with amortized value estimates, 2020. URL <https://arxiv.org/abs/1912.02807>.
- K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition, 2015. URL <https://arxiv.org/abs/1512.03385>.
- D. Hendrycks and K. Gimpel. Gaussian error linear units (gelus), 2023. URL <https://arxiv.org/abs/1606.08415>.
- M. Janner, J. Fu, M. Zhang, and S. Levine. When to trust your model: Model-based policy optimization, 2021. URL <https://arxiv.org/abs/1906.08253>.
- Z. Jiang, T. Zhang, M. Janner, Y. Li, T. Rocktäschel, E. Grefenstette, and Y. Tian. Efficient planning in a compact latent action space, 2023. URL <https://arxiv.org/abs/2208.10291>.
- P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan. Supervised contrastive learning, 2021. URL <https://arxiv.org/abs/2004.11362>.
- M. Lai. Giraffe: Using deep reinforcement learning to play chess, 2015. URL <https://arxiv.org/abs/1509.01549>.
- L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for dimension reduction, 2020. URL <https://arxiv.org/abs/1802.03426>.
- T. M. Moerland, J. Broekens, A. Plaat, and C. M. Jonker. Model-based reinforcement learning: A survey, 2022. URL <https://arxiv.org/abs/2006.16712>.
- Y. Nasu. Efficiently updatable neural-network-based evaluation functions for computer shogi, 2018. Accessed: 2024-04-25.
- G.-C. Pascutto and G. Linscott. Leelachesszero, 2018. URL <https://lczero.org>. Accessed: 2024-04-25.
- A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala. Pytorch: An imperative style, high-performance deep learning library, 2019. URL <https://arxiv.org/abs/1912.01703>.
- M. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley Series in Probability and Statistics. Wiley, 2014. ISBN 9781118625873. URL <https://books.google.com/books?id=VvBjBAAAQBAJ>.
- A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision, 2021. URL <https://arxiv.org/abs/2103.00020>.
- T. Romstad, M. Costalba, J. Kiiski, G. Linscott, Y. Nasu, M. Isozaki, and H. Noda. Stockfish, 2008. URL <https://stockfishchess.org>. Accessed: 2025-04-22.

- A. Ruoss, G. Delétang, S. Medapati, J. Grau-Moya, L. K. Wenliang, E. Catt, J. Reid, C. A. Lewis, J. Veness, and T. Genewein. Amortized planning with large-scale transformers: A case study on chess, 2024. URL <https://arxiv.org/abs/2402.04494>.
- S. Russell and P. Norvig. *Artificial Intelligence, A modern approach*. 1995.
- J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, T. Lillicrap, and D. Silver. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, Dec. 2020. ISSN 1476-4687. doi: 10.1038/s41586-020-03051-4. URL <http://dx.doi.org/10.1038/s41586-020-03051-4>.
- C. E. Shannon. XXII. programming a computer for playing chess. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 41(314):256–275, 1950.
- D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis. Mastering chess and shogi by self-play with a general reinforcement learning algorithm, 2017. URL <https://arxiv.org/abs/1712.01815>.
- N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.*, 15(1):1929–1958, Jan. 2014. ISSN 1532-4435.
- R. S. Sutton. Dyna, an integrated architecture for learning, planning, and reacting. *ACM Sigart Bulletin*, 2(4):160–163, 1991.
- R. S. Sutton and A. G. Barto. *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge, 1998.
- J. v. Neumann. Zur theorie der gesellschaftsspiele. *Mathematische annalen*, 100(1):295–320, 1928.
- A. van den Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding, 2019. URL <https://arxiv.org/abs/1807.03748>.
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need, 2023. URL <https://arxiv.org/abs/1706.03762>.
- A. L. Zobrist. A new hashing method with application for game playing. *ICGA Journal*, 13(2): 69–73, 1990.

A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models were enlisted solely for ancillary tasks, among them formatting and the placement of figures and tables. They were in no wise employed to design experiments, to analyze results, or to generate technical content. All methodological and experimental contributions were conceived and verified exclusively by the authors.

A.2 TRAINING SETUP

We train both SOLIS configurations (Mini and Base) by means of supervised contrastive learning upon 5M ChessBench positions, employing cosine similarity over ℓ_2 -normalized embeddings together with SGD with momentum, on six 48 GB L40S GPUs, for 400k steps at a batch size of 128.

Of these settings, τ and δ are the most consequential. The temperature $\tau = 0.07$ accords with prior contrastive work (Radford et al., 2021; Girdhar et al., 2023), wherein smaller values sharpen the similarity scores while larger ones oversmooth the gradients. We trained with $\tau \in \{0.05, 0.07, 0.1\}$ and ascertained that 0.07 converged most reliably across model sizes. The margin $\delta = 0.05$ strikes

a balance between the collapse of an excessive number of positions into a single neighborhood (attendant upon large δ) and the fragmentation of the space (attendant upon small δ); it moreover reflects the noise inherent in Stockfish evaluations, inasmuch as sub-0.05 differences are seldom meaningful in practice.

Table 3: SOLIS training hyperparameters.

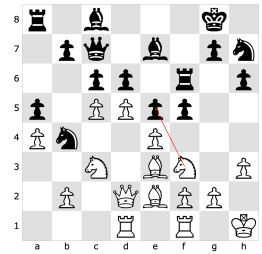
Attribute	Value
Loss	SupCon
τ	0.07
δ (positive margin)	0.05
Positives per anchor	5
Optimizer	SGD
Momentum	0.9
Learning rate	0.05
Batch size	128
Steps	400k
Dropout	0.10
Activation	GELU
Similarity	cosine
Embedding norm	ℓ_2
Token length	77

A.3 STOCKFISH SETUP

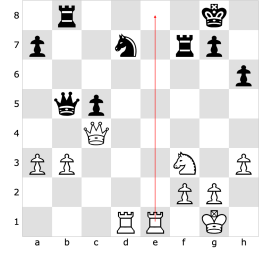
For all experiments we employ the official Linux release of Stockfish 16 (Romstad et al., 2008), configured with a hash size of 32 MB and default threading. Rating caps are imposed by way of its `UCI_LimitStrength` and `UCI_Elo` settings, while depths are governed analogously through the `depth` parameter. To ascertain the number of nodes searched for a given move, we set the engine information level to `INFO_ALL` and extract the `nodes` attribute therefrom.

A.4 REPRESENTATIVE GAMES: STOCKFISH (CAP) VS. SOLIS (DEPTH, WIDTH)

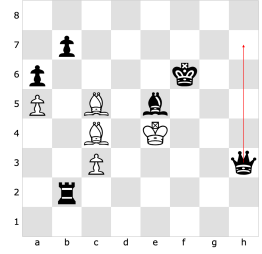
We furnish sample games between SOLIS and Stockfish, wherein SOLIS is confined to search depth S and width W while Stockfish enjoys unbounded search subject to a rating cap. Games *A*, *B*, and *E* exhibit SOLIS' tactical acuity and long-term planning, whereas Games *C* and *D* constitute representative illustrations of its potential frailties. The left portion of each table encodes the moves of the game in Portable Game Notation (PGN) format, while the right portion displays a critical juncture in the game.

1. d4 Nf6 2. c4 d6 3. Nc3 e5 4. Nf3 Qe7 5. e4 c6 6. d5 Qc7 7. Be2 h6 8. a4 a5 9. O-O Be7 10. h3 Na6 11. Be3 Nb4 12. Kh1 Nh7 13. Qd2 O-O 14. Rad1 f5 15. c5 Rf6 16. Nxe5 f4 17. Bxf4 Rxf4 18. Qxf4 dxe5 19. d6 exf4 20. Bc4+ Kf8 21. dxc7 Ke8 22. e5 Ng5 23. Rd6 Bxd6 24. exd6 Bd7 25. Re1+ Kf8 26. Re7 Bf5 27. h4 Nd5 28. Nxd5 cxd5 29. Bxd5 Be6 30. Bxe6 f3 31. hxg5 b6 32. Bh3 hxg5 33. Re5 Rc8 34. Rf5+ Kg8 35. Rf7 g4 36. cxb6 g6 37. d7 fxg2+ 38. Kh2 Rf8 39. Bxg2 Kxf7 40. Bc6 Ke7 41. Kg3 Rf3+ 42. Kh2 Rd3 43. b7 Rh3+ 44. Kg2 Re3 45. d8=Q+ Kf7 46. b8=Q Rg3+ 47. Kh2 Rh3+ 48. Kg1 Rh7 49. Qe8+ Kf6 50. Qbd8+ Kf5 51. Be4+ Kf4 52. Qf6# 1-0	
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Game A — SOLIS ($S=5$, $W=3$) vs. SF 2350, SOLIS–White. Result: 1–0. SOLIS sacrifices a knight so as to secure an advantage in the middlegame.

1. d4 d5 2. e3 c6 3. Bd3 h6 4. h3 Nf6 5. Nf3 e6 6. O-O b6 7. c4 Ba6 8. b3 Nbd7 9. Bb2 Bb4 10. a3 Be7 11. Nbd2 O-O 12. Re1 c5 13. Qe2 Qc7 14. e4 Nxe4 15. Nxe4 dxe4 16. Bxe4 Rab8 17. d5 Bf6 18. dxe6 Bxb2 19. exf7+ Rxf7 20. Qxb2 Bb7 21. Bxb7 Qxb7 22. Rad1 b5 23. Qe2 bxc4 24. Qxc4 Qb5 25. Re8+ Nf8 26. Rxb8 Qxb8 27. Rd5 Qf4 28. Qxf4 Rxf4 29. Rxc5 Ne6 30. Rc6 Nd4 31. Nxd4 a5 32. Rc4 Rf8 33. Nc6 Kh8 34. Nxa5 Ra8 35. b4 Rd8 36. g3 Rd3 37. a4 g5 38. b5 Rd2 39. b6 Rd5 40. b7 Rxa5 41. Rc8+ Kh7 42. b8=Q Rd5 43. Rc7+ Kg6 44. Qg8+ Kf5 45. Qxd5+ Kg6 46. Qf7# 1-0	
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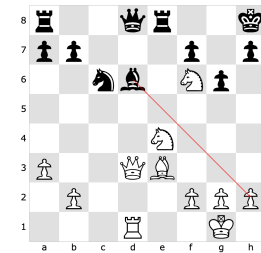
Game B — SOLIS ($S=5$, $W=3$) vs. SF 2300, SOLIS–White. Result: 1–0. SOLIS sacrifices a rook in order to win Stockfish's queen, and thereafter converts the resulting advantage into a win.

1. e3 Nf6 2. Nc3 d5 3. Nf3 c5 4. d4 e6 5. g3 a6 6. a4 Nc6 7. Ne2 Bd6 8. Bg2 O-O 9. h3 Re8 10. c3 Qc7 11. Nd2 e5 12. Nb3 c4 13. dxe5 Nxe5 14. Nbd4 Nd3+ 15. Kf1 Nc5 16. Bf3 Bd7 17. Kg2 Be5 18. a5 Rad8 19. Qc2 g6 20. Bd2 h5 21. Be1 h4 22. gxh4 Kg7 23. Ng3 Rh8 24. h5 Rdg8 25. hxg6 fxg6 26. Be2 Qd6 27. Qd1 Qe7 28. f3 Bd6 29. Nf1 Rh5 30. h4 Rgh8 31. Bf2 R5h7 32. Qe1 Bb8 33. Bg3 Ba7 34. Bf2 Nh5 35. Ng3 Bb8 36. f4 Nf6 37. h5 Rh6 38. hxg6 Rxh1 39. Qxh1 Qe8 40. Ndf5+ Bxf5 41. Nxf5+ Kxg6 42. Ne7+ Kf7 43. e4 Rxh1 44. Rxh1 Qxe7 45. Bxc5 Qxe4+ 46. Bf3 Qg6+ 47. Kf1 Bxf4 48. b4 cxb3 49. Rh8 Ke6 50. Rf8 b2 51. Rxf6+ Kxf6 52. Bxd5 b1=R+ 53. Ke2 Rb2+ 54. Kf3 Be5 55. Bc4 Qh5+ 56. Ke3 Qh3+ 57. Ke4 Qh7+ 58. Kf3 Qh3+ 59. Ke4 Qh7+ 60. Kf3 Qh3+ 61. Ke4 Qh7+ 62. Kf3 Qh3+ 63. Ke4 Qh7+ 64. Kf3 Qh3+ 65. Ke4 1/2-1/2	
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Game C — SF 2200 vs. SOLIS ($S=5$, $W=3$), SOLIS–Black. Result: 1/2–1/2. SOLIS emerges into a winning endgame position, yet settles for a draw on account of its blindness to threefold repetition.

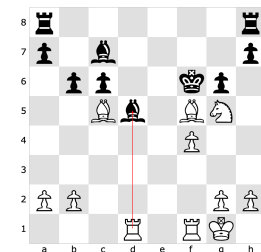
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1. d4 d5 2. c4 dxc4 3. e3 e5 4. Bxc4 exd4 5. exd4
Nf6 6. Nc3 Bd6 7. Qe2+ Qe7 8. Be3 O-O 9. Nf3 c6 10.
O-O Be6 11. Rfd1 Re8 12. d5 cxd5 13. Bxd5 Nc6 14.
a3 Nxd5 15. Nxd5 Qd8 16. Qd3 Bg4 17. Ng5 g6 18. Ne4
Bxd1 19. Ndf6+ Kh8 20. Rxd1 Bxh2+ 21. Kxh2 Qxd3 22.
Rxd3 Red8 23. Nd7 Kg7 24. f3 h6 25. Rd6 f5 26. Nec5
Re8 27. Bd2 Re2 28. Kh1 Kf7 29. b3 Ke7 30. Nxb7 Rc8
31. Ndc5 Kf7 32. Kg1 Kg8 33. Rxg6+ Kf7 34. Rd6 Ne7
35. Rd7 Rg8 36. Nd6+ Kg6 37. Nc4 Kh5 38. Ne3 f4 39.
Ne6 fxe3 40. Nf4+ Kg5 41. Nxe2 exd2 42. Kf2 Kf6 43.
Nc3 Ke6 44. Rd4 Rc8 45. Ne4 d1=N+ 46. Rxd1 Rc2+ 47.
Kg1 Nf5 48. Rd2 Rxd2 49. Nxd2 Ke5 50. Kh2 Kf4 51.
b4 Ke3 52. Ne4 Nd4 53. a4 Nc2 54. b5 Nd4 55. Nd6
Ne6 56. a5 Kd3 57. g4 Kd4 58. b6 axb6 59. axb6 Nc5
60. Kg3 Kd5 61. b7 Na6 62. Nf7 Kc6 63. Nxh6 Kxb7
64. f4 Kc7 65. Kh4 Kd7 66. Kh5 Nc5 67. f5 Kd6 68.
f6 Ne6 69. Nf7+ Kc6 70. Kh6 Kd5 71. g5 Nc5 72. Kg6
Ne6 73. Kh5 Kd4 74. g6 Kc4 75. Ne5+ Kd5 76. Ng4 Ke4
77. Kh6 Kf5 78. g7 Nxg7 79. Kxg7 Kxg4 80. Kf7 Kf5
81. Ke7 Kg5 82. f7 Kf4 83. Kf6 Kf3 84. Ke5 Kg3 85.
Ke4 Kf2 86. Kd3 Kg1 87. Ke3 Kh2 88. f8=Q Kg3 89.
Qg8+ Kh3 90. Kf4 Kh2 91. Kf3 Kh3 92. Qg3# 1-0



Game D — SF 2500 vs. SOLIS ($S=5$, $W=3$), SOLIS–Black. Result: 1-0 SOLIS commits a short-sighted piece sacrifice, thereby incurring a long-term material disadvantage.

1. d4 d5 2. e3 Nd7 3. c4 e5 4. dxe5 dxc4 5. Nf3
Qe7 6. Bxc4 Nxe5 7. Nxe5 Qxe5 8. Qb3 Qh5 9. e4
Nf6 10. O-O Bd6 11. Qb5+ c6 12. Qxh5 Nxh5 13. Nc3
Be5 14. f4 Bc7 15. Be3 g6 16. Bb3 Be6 17. Rad1 Ng7
18. Bc2 f5 19. Bc5 Kf7 20. exf5 Nxf5 21. Ne4 b6
22. Ng5+ Kf6 23. Bxf5 Bd5 24. Rxd5 cxd5 25. Bd4+
Kxf5 26. h3 Bxf4 27. Bxh8 Kxg5 28. Bg7 Rg8 29. Bd4
Re8 30. Rf3 Re1+ 31. Kf2 Re6 32. g3 Bd2 33. Rf7
Re4 34. Bf6+ Kh6 35. g4 a6 36. Kg3 d4 37. h4 Be1+
38. Kh3 Re3+ 39. Kg2 Bxh4 40. Bxh4 a5 41. Bf6 a4
42. Rd7 Rd3 43. Rd5 b5 44. Rxb5 a3 45. Re5 Rd2+
46. Kf3 axb2 47. g5+ Kh5 48. Rb5 Rd3+ 49. Kf4 h6
50. gxh6+ Kxh6 51. Rxb2 g5+ 52. Kg4 Rd1 53. Bxg5+
Kg6 54. Rb6+ Kg7 55. Kf5 Kf7 56. Rb7+ Ke8 57. a4
Ra1 58. Re7+ Kf8 59. Rh7 Rxa4 60. Bh6+ Kg8 61. Re7
Rc4 62. Kg5 Rc5+ 63. Kg4 Ra5 64. Rg7+ Kh8 65. Re7
d3 66. Rd7 d2 67. Bxd2 Rc5 68. Bf4 Kg8 69. Be3 Rb5
70. Bh6 Re5 71. Kf4 Rc5 72. Ke4 Rb5 73. Kf4 Rb4+
74. Ke5 Rb1 75. Kf6 Rb2 76. Kg6 Rb6+ 77. Kh5 Rb2
78. Bg5 Rb6 79. Bh6 Re6 80. Kg5 Ra6 81. Rg7+ Kh8
82. Re7 Ra5+ 83. Kg6 Ra6+ 84. Kh5 Kg8 85. Rb7 Rc6
86. Rd7 Rc5+ 87. Bg5 Rc2 88. Kg6 Rc6+ 89. Bf6 Re6
90. Rd8+ Re8 91. Rxe8# 1-0

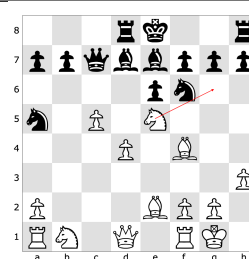


Game E — SOLIS ($S=5$, $W=3$) vs. Stockfish 2500 , SOLIS–White. Result: 1-0 SOLIS undertakes an exchange sacrifice in return for long-term compensation, and thereafter converts the advantage into a win.

A.5 EQUAL-DEPTH MATCHES: SOLIS D vs. STOCKFISH D

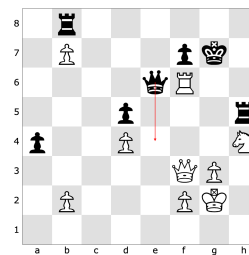
We include several illustrative games in which SOLIS and Stockfish search to the identical depth D . These examples serve to underscore the manner in which SOLIS' latent planning competes head-to-head absent any rating cap upon Stockfish. The left portion of each table encodes the moves of the game in Portable Game Notation (PGN) format, while the right portion displays a critical juncture in the game.

1. d4 d5 2. e3 c5 3. Nf3 Nf6 4. Be2 e6 5. O-O Nc6
6. b3 cxd4 7. exd4 Be7 8. c3 Qc7 9. h3 Bd7 10. c4
dxc4 11. bxc4 Na5 12. Ne5 Rd8 13. Bf4 Bd6 14. c5
Be7 15. Ng6 Qc8 16. Nxh8 Bc6 17. Nc3 Kf8 18. Rc1
Qa8 19. Bc7 b5 20. Bxa5 Bxg2 21. d5 Bxh3 22. Qd3
Bf5 23. Qg3 b4 24. Rfd1 Bxc5 25. Qc7 Be7 26. Nb5
Nxd5 27. Rxd5 Qxd5 28. Bxb4 Qd7 29. Qxd7 Re8 30.
Rc8 g5 31. Qxe7+ Kg7 32. Qxf7+ Kh6 33. Bf8+ Rxf8 34.
Qxf8# 1-0



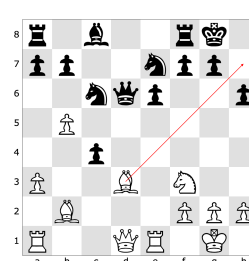
Game F — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 SOLIS sacrifices a knight during the middlegame in exchange for a concrete tactical advantage.

1. d4 d5 2. e3 c5 3. Nf3 e6 4. Be2 cxd4 5. exd4
Bd6 6. O-O Nf6 7. Bg5 Nbd7 8. Nbd2 h6 9. Bh4 O-O
10. Re1 a5 11. Bf1 Bf4 12. c3 Qb6 13. Qc2 Qc7 14.
a4 b6 15. Bb5 Ba6 16. Bxa6 Rxa6 17. Qd3 Ra7 18. Re2
Qc8 19. Ree1 Nh5 20. Nf1 Nh6 21. N1d2 Nh5 22. Nf1
Bc7 23. Qd1 Nh6 24. Qd3 Nh5 25. Qd1 Nh6 26. Qd3
Rb7 27. Qb5 Ra7 28. N1d2 Nh5 29. h3 g5 30. Rac1
gxh4 31. c4 Nf4 32. Rc3 Qd8 33. g3 Nxh3+ 34. Kg2
Ng5 35. cxd5 h3+ 36. Kg1 h2+ 37. Kxh2 exd5 38. Rc6
Nf6 39. Nxc5 hxg5 40. Kg2 Re8 41. Rh1 Bd6 42. Nf3
Kg7 43. Nxc5 Rc7 44. Rxd6 Qxd6 45. Qf1 b5 46. axb5
Rcc8 47. Rh4 Rh8 48. Nf3 Ne4 49. Rf4 Rh5 50. Qe2
Qe6 51. Nh4 Rch8 52. b6 Nf6 53. Qf3 a4 54. b7 Rb8
55. Rxf6 Qe4 56. Rxf7+ Kh8 57. Qxe4 dxe4 58. Rc7
Rg5 59. Rc8+ Rg8 60. Rc7 Rg5 61. Rc8+ Rg8 62. Ng6+
Kg7 63. Ne7 Rf8 64. Rc7 Rf7 65. d5 Rbf8 66. Nf5+
Kg6 67. Rxf7 Kxf7 68. Nd6+ Kg6 69. Nc8 Kg5 70. b8=Q
Kg4 71. Qc7 Rf5 72. Nd6 e3 73. Qg7+ Rg5 74. f3+ Kh5
75. Qh7# 1-0



Game G — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 Stockfish steadily accrues an advantage but subsequently blunders during the late middlegame; SOLIS capitalizes upon this error and prevails in the endgame.

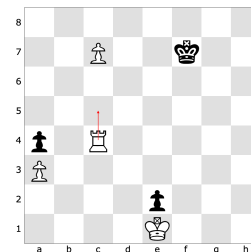
1. d4 c6 2. e4 d5 3. Nc3 Nf6 4. e5 Ng8 5. Bd3 e6
6. Nf3 c5 7. dxc5 Bxc5 8. O-O Nc6 9. a3 h6 10. b4
Bb6 11. Nb5 Bc7 12. Re1 Bb8 13. c4 Nge7 14. Nd6+
Bxd6 15. exd6 Qxd6 16. Bb2 O-O 17. b5 dxc4 18. Bh7+
Kh8 19. Qxd6 f6 20. Be4 c3 21. Bxc3 e5 22. bxc6 Bf5
23. Qxe7 Bxe4 24. Rxe4 bxc6 25. Nxe5 Rfe8 26. Nf7+
Kg8 27. Nxh6+ Kh7 28. Rae1 Rad8 29. Ba5 Rd4 30. g4
Rxe4 31. Rxe4 Rb8 32. Qxa7 Rd8 33. Rf4 Rd3 34. g5
fxg5 35. Rd4 Rxa3 36. Kf1 Rb3 37. Rb4 Rc3 38. h4
gxh4 39. Rd4 h3 40. Qe7 h2 41. Qe4+ Kxh6 42. Rd6+
Kg5 43. Rg6+ Kh5 44. Qg4# 1-0



Game H — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1-0 SOLIS sacrifices a bishop in return for a discovered attack upon Stockfish's queen, and thereafter converts the advantage into a win.

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1. d4 d5 2. e3 c6 3. c4 e6 4. Nf3 Nf6 5. Be2 c5
6. O-O Nc6 7. b3 a6 8. Bb2 Be7 9. a3 cxd4 10. exd4
O-O 11. c5 b6 12. b4 bxc5 13. dxc5 Bb7 14. Nbd2 a5
15. b5 Bxc5 16. Rc1 Qb6 17. bxc6 Ng4 18. Rxc5 Ba6
19. Rc2 Nxf2 20. Rxf2 Bxe2 21. Qxe2 a4 22. c7 Rac8
23. Bd4 Qb5 24. Qxb5 Rfe8 25. Qb8 h6 26. Ne5 g5 27.
h4 Rf8 28. Ne4 dxe4 29. Rxf7 Rxb8 30. Rxf8+ Rxf8 31.
hxcg5 Rc8 32. gxh6 e3 33. Bxe3 Kh7 34. Rc4 Kg8 35.
Kh2 Kh7 36. g4 Rg8 37. g5 Ra8 38. Rg4 Kh8 39. Ba7
Rf8 40. Bb8 Rf2+ 41. Kg1 Rc2 42. Rc4 Rb2 43. Ba7
Rb1+ 44. Kg2 Kh7 45. Bf2 Rb2 46. g6+ Kxh6 47. Kh2
Rxf2+ 48. Kg3 Rg2+ 49. Kxg2 Kg7 50. Rc1 Kf6 51. Kh2
Kg7 52. Nd7 e5 53. Nf6 Kxf6 54. Rg1 Kg7 55. Rf1 Kh6
56. Rg1 Kg7 57. Rf1 Kxg6 58. Rc1 Kg7 59. Rc4 Kf7
60. Kg2 Ke7 61. Kf2 Kd6 62. Ke1 Kd5 63. Rc1 Ke4 64.
Ke2 Kd4 65. Rc2 e4 66. Rc1 Kd5 67. Ke1 Ke5 68. Rc4
Kd5 69. Rc2 Kd6 70. Rc1 Ke7 71. Rc4 Kf7 72. Rc5 e3
73. Kf1 Ke7 74. Rc4 Kf7 75. Kg2 e2 76. Kf2 Ke7 77.
Ke1 Kf7 78. Rc5 Kg7 79. Rc2 Kf6 80. Rc4 Kf7 81. Rc5
Kg6 82. Kd2 Kf7 83. Ke1 Ke7 84. Rc4 Kf7 85. Rc5 Kg6
86. Rc4 Kf7 87. Rc5 1/2-1/2



Game I — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1/2-1/2 SOLIS struggles to plan within a position in which every trajectory is winning.

A.6 SOLIS VS. STOCKFISH RESULTS

In order to substantiate the reproducibility of our Elo estimates, we provide the full match outcomes between SOLIS and Stockfish, wherein SOLIS searches at depths $S \in \{2, 3, 4, 5\}$. Each table is indexed by Stockfish’s Elo cap along the rows and by the SOLIS search width W along the columns. The entries are rendered as win–loss–draw counts from the respective matchups, with the total match points supplied parenthetically. Separate tables are reported for the Base and Mini models and, within each, for both the unanchored and the anchored scoring regimes.

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 6: Base model contra Stockfish: per-depth blocks obtained under unanchored planning, reported as W–D–L (score) out of 100.

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Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	32-32-36 (48)	27-45-28 (50)	-	-	1350	-	-	-	-
1400	30-41-29 (50)	33-27-40 (46)	-	-	1400	-	-	-	-
1450	29-44-27 (51)	33-40-27 (53)	-	-	1450	-	-	-	-
1500	31-38-31 (50)	31-39-30 (50)	-	-	1500	29-38-33 (48)	28-43-29 (50)	-	-
1550	38-35-27 (56)	23-50-27 (48)	-	-	1550	41-36-23 (59)	36-37-27 (54)	-	-
1600	34-30-36 (49)	29-37-34 (48)	-	-	1600	37-30-33 (52)	37-30-33 (52)	-	-
1650	24-45-31 (46)	28-36-36 (46)	-	-	1650	32-38-30 (51)	30-40-30 (50)	-	-
1700	-	-	-	-	1700	28-37-35 (46)	21-43-36 (42)	-	-
1750	-	-	-	-	1750	36-33-31 (52)	28-40-32 (48)	-	-
1800	-	-	37-31-32 (52)	37-36-27 (55)	1800	39-24-37 (51)	29-35-36 (46)	-	-
1850	-	-	34-24-42 (46)	34-31-35 (50)	1850	-	-	-	-
1900	-	-	40-28-32 (54)	40-29-31 (54)	1900	-	-	-	-
1950	-	-	39-29-32 (54)	39-29-32 (54)	1950	-	-	-	-
2000	-	-	38-34-28 (55)	42-28-30 (56)	2000	-	-	34-35-31 (52)	38-21-41 (48)
2050	-	-	41-27-32 (54)	40-26-34 (53)	2050	-	-	38-34-28 (55)	34-33-33 (50)
2100	-	-	35-30-35 (50)	36-26-38 (49)	2100	-	-	33-34-33 (50)	34-29-37 (48)
2150	-	-	-	-	2150	-	-	31-35-34 (48)	38-25-37 (50)
2200	-	-	-	-	2200	-	-	38-18-44 (47)	31-26-43 (44)
2250	-	-	-	-	2250	-	-	36-23-41 (48)	45-26-29 (58)
2300	-	-	-	-	2300	-	-	40-22-38 (51)	34-26-40 (47)
2350	-	-	-	-	2350	-	-	-	-
2400	-	-	-	-	2400	-	-	-	-
2500	-	-	-	-	2500	-	-	-	-
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	-	-	-	-	1350	-	-	-	-
1400	-	-	-	-	1400	-	-	-	-
1450	-	-	-	-	1450	-	-	-	-
1500	-	-	-	-	1500	-	-	-	-
1550	-	-	-	-	1550	-	-	-	-
1600	36-42-22 (57)	35-31-34 (50)	-	-	1600	-	-	-	-
1650	29-40-31 (49)	36-33-31 (52)	-	-	1650	-	-	-	-
1700	33-34-33 (50)	36-32-32 (52)	-	-	1700	39-30-31 (54)	34-34-32 (51)	-	-
1750	43-26-31 (56)	28-38-34 (47)	-	-	1750	37-29-34 (52)	32-30-38 (47)	-	-
1800	38-24-38 (50)	30-37-33 (48)	-	-	1800	42-20-38 (52)	33-31-36 (48)	-	-
1850	49-23-28 (60)	29-33-38 (46)	-	-	1850	36-24-40 (48)	43-28-29 (57)	-	-
1900	-	-	-	-	1900	45-26-29 (58)	46-31-23 (62)	-	-
1950	-	-	-	-	1950	40-22-38 (51)	37-35-28 (54)	-	-
2000	-	-	-	-	2000	-	-	-	-
2050	-	-	-	-	2050	-	-	-	-
2100	-	-	-	-	2100	-	-	-	-
2150	-	-	31-31-38 (46)	44-21-35 (54)	2150	-	-	-	-
2200	-	-	35-29-36 (50)	48-27-25 (62)	2200	-	-	27-37-36 (46)	33-38-29 (52)
2250	-	-	37-31-32 (52)	37-28-35 (51)	2250	-	-	30-33-37 (46)	38-31-31 (54)
2300	-	-	27-37-36 (46)	36-34-30 (53)	2300	-	-	38-37-25 (56)	33-39-28 (52)
2350	-	-	22-42-36 (43)	37-35-28 (54)	2350	-	-	32-29-39 (46)	29-36-35 (47)
2400	-	-	36-29-35 (50)	37-26-37 (50)	2400	-	-	30-26-44 (43)	30-32-38 (46)
2500	-	-	-	-	2500	-	-	26-34-40 (43)	43-25-32 (56)

Figure 7: Mini model contra Stockfish: per-depth blocks obtained under unanchored planning, reported as W-D-L (score) out of 100.

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Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 8: Base model contra Stockfish: per-depth blocks obtained under anchored planning, reported as W–D–L (score) out of 100.

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Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	36–32–32 (52)	39–30–31 (54)	–	–	1350	–	–	–	–
1400	23–49–28 (48)	30–32–38 (46)	–	–	1400	–	–	–	–
1450	33–37–30 (52)	24–47–29 (48)	–	–	1450	–	–	–	–
1500	34–38–28 (53)	25–40–35 (45)	–	–	1500	29–40–31 (49)	35–33–32 (52)	–	–
1550	28–38–34 (47)	40–39–21 (60)	–	–	1550	34–35–31 (52)	37–29–34 (52)	–	–
1600	23–47–30 (46)	28–42–30 (49)	–	–	1600	32–36–32 (50)	32–39–29 (52)	–	–
1650	35–29–36 (50)	38–37–25 (56)	–	–	1650	34–30–36 (49)	37–33–30 (54)	–	–
1700	–	–	–	–	1700	36–30–34 (51)	34–35–31 (52)	–	–
1750	–	–	–	–	1750	34–25–41 (46)	34–30–36 (49)	–	–
1800	–	–	35–37–28 (54)	40–28–32 (54)	1800	32–29–39 (46)	32–31–37 (48)	–	–
1850	–	–	32–32–36 (48)	44–29–27 (58)	1850	–	–	–	–
1900	–	–	37–30–33 (52)	45–21–34 (56)	1900	–	–	–	–
1950	–	–	43–30–27 (58)	36–29–35 (50)	1950	–	–	–	–
2000	–	–	35–31–34 (50)	43–27–30 (56)	2000	–	–	46–32–22 (62)	38–27–35 (52)
2050	–	–	37–29–34 (52)	45–27–28 (58)	2050	–	–	37–27–36 (50)	39–26–35 (52)
2100	–	–	36–27–37 (50)	42–34–24 (59)	2100	–	–	29–34–37 (46)	48–17–35 (56)
2150	–	–	–	–	2150	–	–	37–33–30 (54)	39–25–36 (52)
2200	–	–	–	–	2200	–	–	37–31–32 (52)	41–23–36 (52)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	44–20–36 (54)
2300	–	–	–	–	2300	–	–	30–43–27 (52)	38–26–36 (51)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	26–45–29 (48)	31–33–36 (48)	–	–	1600	–	–	–	–
1650	37–39–24 (56)	28–32–40 (44)	–	–	1650	–	–	–	–
1700	23–39–38 (42)	29–41–30 (50)	–	–	1700	37–36–27 (55)	34–29–37 (48)	–	–
1750	36–29–35 (50)	38–30–32 (53)	–	–	1750	42–23–35 (54)	37–33–30 (54)	–	–
1800	35–24–41 (47)	33–36–31 (51)	–	–	1800	36–28–36 (50)	31–35–34 (48)	–	–
1850	41–21–38 (52)	30–28–42 (44)	–	–	1850	38–21–41 (48)	33–32–35 (49)	–	–
1900	–	–	–	–	1900	37–27–36 (50)	37–37–26 (56)	–	–
1950	–	–	–	–	1950	43–18–39 (52)	38–24–38 (50)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	39–23–38 (50)	32–30–38 (47)	2150	–	–	–	–
2200	–	–	33–40–27 (53)	38–25–37 (50)	2200	–	–	30–40–30 (50)	32–34–34 (49)
2250	–	–	36–31–33 (52)	30–27–43 (44)	2250	–	–	29–43–28 (50)	49–26–25 (62)
2300	–	–	32–33–35 (48)	37–28–35 (51)	2300	–	–	31–30–39 (46)	30–35–35 (48)
2350	–	–	35–34–31 (52)	40–25–35 (52)	2350	–	–	26–32–42 (42)	40–34–26 (57)
2400	–	–	28–26–46 (41)	26–38–36 (45)	2400	–	–	39–32–29 (55)	31–32–37 (47)
2500	–	–	–	–	2500	–	–	30–29–41 (44)	28–41–31 (48)

Figure 9: Mini model contra Stockfish: per-depth blocks obtained under anchored planning, reported as W–D–L (score) out of 100.